

Solution Requirements

Date	14 July 2026
Team ID	
Project Name	SMART LENDER
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Provide credit officer login with email and password.	High
2	Authorization levels	Define roles: Applicant (read-only predictions), Credit Officer (full access).	Medium
3	External interfaces	Connect to external loan history database via secure APIs.	Medium

4	Transactions processing	Validate and process applicant demographic and credit score inputs.	High
5	Reporting	Generate visual summary reports of approval rates and model accuracies.	Medium
6	Business rules, etc.	If credit history is 0, reject loan unless income is exceptionally high.	High
7	Compliance to laws or regulations	Adhere to data privacy and protection regulations for personal financial data.	High
8	<i>Other</i>	Model must predict with XGBoost classifier and compare with KNN.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	Model inference and page rendering takes < 1 second.	Inference < 1s
2	Scalability	Handles up to 1,000 concurrent prediction requests.	Supports 1,000 users
3	Security & Data Privacy	Secure storage of applicant credentials and inputs.	Data encryption
4	Reliability & Availability	Ensure 99% uptime of the Flask server.	99% uptime
5	Usability & Accessibility	Clean, responsive design with clear text for bank staff.	Responsive Web UI
6	<i>Other</i>	Easy model retrainability when new historical data arrives.	Model modularity